

Lecture: Product Lifecycle Mastery

1. Idea Validation — "Is this worth building?"

Before anything is built, you need to confirm that a real problem exists and that real people care about solving it. Skipping this is the number one reason products fail.

The core question: Does this problem exist, and are people frustrated enough to pay for a solution?

Practical example: Imagine someone at X3 Lab says *"let's build an app that helps students track their SIWES log automatically."* Before writing one line of code, you should ask: Do students actually struggle with this? Are supervisors complaining about incomplete logs? Could you solve it with a WhatsApp group instead?

Validation methods:

- Talk to at least 5–10 real users before building anything
- Run a simple survey (Google Forms costs nothing)
- Build a fake landing page and see if people sign up
- Look for workarounds — if students are already using Excel or WhatsApp to track logs, that's proof the problem exists

Practical task I'm setting you: Pick any product you use daily (could be Jumia, a school portal, anything). Write down the exact problem it was solving before it existed. That's idea validation working backwards.

2. MVP — Minimum Viable Product

An MVP is the simplest version of your product that still delivers value. It is not a broken product. It is a focused one.

Think of it this way: If you want to build a car, your MVP is not half a car with no wheels. Your MVP is a bicycle — it solves the transportation problem while you figure out whether people actually want to go faster.

Real example: When Instagram launched, it only did one thing — apply filters to photos and share them. No reels, no stories, no shopping. Just filters. That was the MVP. They validated that people loved it, then built everything else.

For your X3 Lab assignment: Look at whatever initiative you're studying and ask — what was the very first version? What did it do and what did it deliberately leave out? That's the MVP.

Common mistakes:

- Adding too many features "just in case" — this wastes time and money

- Confusing MVP with prototype — a prototype tests design, an MVP tests the market
- Waiting until it's "perfect" to release — done is better than perfect at this stage

Practical task: Write a one-paragraph MVP description for a student logbook app. What does version 1 do? What does it not do yet?

3. Product-Market Fit — "Did we get it right?"

Product-market fit (PMF) is the moment when your product clicks with the market. Users don't just use it — they need it. They tell others. They're upset when it's down.

The simplest test: Ask your users *"How would you feel if you could no longer use this product?"* If more than 40% say "very disappointed," you likely have product-market fit. This is called the Sean Ellis test.

Real example: WhatsApp knew they had PMF when people in Nigeria, India, and Brazil were paying \$1/year for it despite having free SMS. The product fit the market's need so well that people paid for what was free elsewhere.

Signs you DON'T have PMF yet:

- Users sign up but don't come back
- You have to constantly convince people to use it
- People say "it's nice" but don't recommend it to anyone

For your assignment: When studying the X3 Lab initiative — did it achieve PMF? How do you know? What evidence exists (user numbers, feedback, repeat usage)?

4. Iteration Cycles — "How do we improve?"

No product is right the first time. Iteration is the process of releasing, measuring, learning, and improving — repeatedly.

The loop looks like this:

Build → Release → Measure → Learn → Build again

Real example: Think about how your phone's apps update constantly. Every update is the result of one iteration cycle. Developers watched how you used the app, saw where you got confused or stopped, and fixed it.

Key tools used in iteration:

- User feedback (surveys, interviews, complaints)
- Analytics (which features do people actually use?)

- A/B testing (show version A to half your users, version B to the other half — see which works better)

What to look for in your X3 Lab study: How many versions did the initiative go through? What changed between version 1 and the current version? What triggered each change?

Practical task: Pick any app on your phone. Look at its update history in the Play Store or App Store. Pick two updates and write one sentence each explaining what problem that update was likely solving.

5. Growth Stages — "Where is the product in its life?"

Every product goes through five stages. Knowing which stage a product is in tells you what decisions to make.

Stage	What's happening	Key focus
Introduction	Just launched, low awareness	Getting first users
Growth	Users increasing fast	Scaling, fixing bugs
Maturity	Growth slows, competitors appear	Retention, differentiation
Decline	Users dropping	Pivot, improve, or shut down
Renewal (optional)	Product reinvents itself	New market or new form

Real example: BlackBerry phones went through all five stages. Introduction in the 2000s, massive growth with BBM, maturity as smartphones rose, decline when Android and iPhone took over. They attempted renewal by switching to software — with limited success.

Facebook is in maturity — growth has slowed in the West, so they're focused on retention and new markets (hence WhatsApp and Instagram acquisitions).

For your assignment: Map the X3 Lab initiative to one of these stages and justify why. What evidence supports your answer?

Test — Answer These Before Friday

1. What is the difference between an MVP and a prototype? Give a one-sentence answer.
2. You built a student portal. 30 out of 100 users say they'd be "very disappointed" if it shut down. Do you have product-market fit? Why or why not?
3. Name the 5 growth stages in order and place one real Nigerian or African product at each stage.
4. Look at your X3 Lab initiative — write 3 bullet points identifying what lifecycle stage it's in and why.

5. What is one missing system you identified in the initiative, and what lifecycle stage would fixing it belong to?